



Benjamin BILLOT

Researcher at Inria

Inria, Epione team

2004 route des Lucioles, 06560 Sophia Antipolis, FRANCE

 benjamin.billot@inria.fr

 [Benjamin Billot](#)

 [BBillot](#)

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Experience

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|--|--------------------------------|---|
| 2025 - now | Researcher | Epione team, Inria Sophia Antipolis, France |
| Permanent researcher of the Epione team at Inria Sophia Antipolis, where my work focuses on developing machine learning strategies to robustly analyse medical images. | | |
| 2022 - 2024 | Postdoctoral researcher | MIT, USA |
| Equivariant and denoising CNNs to decouple spatial and intensity features for motion tracking in fetal MRI
Advisor: Polina Golland | | |
| 2018 | Data Scientist | Founders Factory, London, UK |
| Developed R-CNN for automated navigation of HTML pages | | |
| Summer 2016 | Research assistant | INSERM, France |
| Physics-based generative models in low data regime: segmentation of cortical microscopies.
Advisors: Thérèse Jay and Marco Pompili | | |

Education

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|---|---------------------------------------|-------------------------------|
| 2022 | PhD in medical image computing | University College London, UK |
| Thesis: Bridging generative models and CNNs for domain-agnostic segmentation of brain MRI
Advisor: Juan Eugenio Iglesias | | |
| 2017 | MSc in biomedical engineering | Imperial College London, UK |
| Thesis: Physics-based generative models in low data regime: segmentation of cortical microscopies.
Advisor: Anil Bharath | | |
| 2016 | Diplôme d'ingénieur | CentraleSupélec, France |
| Project: Cancellation of respiratory motions for accurate chest CT reconstruction
Advisor: Elisabeth Lahalle | | |

Honours and awards

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| 2025 | IdEx Attractivity Package |
| 50,000€ awarded by IdEx Université Côte d'Azur | |
| 2022-2025 | Outstanding reviewer |
| MICCAI 2022, MICCAI 2023, MICCAI 2024, MICCAI 2025 | |
| 2019-2024 | Short-listed for best paper award |
| MIDL 2019, MIDL 2020, MIDL 2024 | |
| 2019 | CMIC platform grant |
| £4,000 awarded for a 6-month visit at MIT as a PhD student | |

Summary of publications

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| 18 | Journal articles | 4 as first author, 4 as second author |
| 20 | Conference articles | 7 as first author, 5 as second author |

Software releases

SynthSeg	Domain-agnostic segmentation of brain scans Python/TensorFlow	GitHub FreeSurfer Matlab	github.com/BBillot/SynthSeg surfer.nmr.mgh.harvard.edu/fswiki/SynthSeg mathworks.com/products/matlab.html
SynthSR	Tool to turn any brain scan into a 1mm T1 scan Python/TensorFlow	GitHub FreeSurfer	github.com/BBillot/SynthSR surfer.nmr.mgh.harvard.edu/fswiki/SynthSR
Hypo_seg	Segmentation of the hypothalamic subunits Python/TensorFlow	GitHub FreeSurfer	github.com/BBillot/hypothalamus_seg surfer.nmr.mgh.harvard.edu/fswiki/HypothalamicSubunits
EquiTrack	Motion Tracking in fetal MRI time series Python/PyTorch	GitHub	github.com/BBillot/EquiTrack

Mentoring

2025 - now	Huiyu Li – Postdoctoral researcher Robust AI detection tool for lung cancer prognosis Co-supervised with Hervé Delingette (EPIONE) and Charles Marquette (IHU Nice)	Inria, France
2025 - now	Alix de Langlais - PhD student 3D modelling of proximal humerus fractures for preoperative planning and intraoperative assistance Co-supervised with Hervé Delingette (EPIONE) and Marc-Olivier Gauci (CHU Nice)	Inria, France
2025 - now	Manasi Kattel - PhD student Deep learning methods for the analysis and exploitation of ultrasound images CIFRE thesis with KOELIS, co-supervised with Nicholas Ayache (EPIONE) and Hervé Delingette (EPIONE)	Inria, France
2025 - now	Federica Facente - PhD student Biomedical models for CT/X-ray image analysis - application to image-guided surgical robotics CIFRE thesis with CARANX MEDICAL, co-supervised with Nicholas Ayache (EPIONE) and Hervé Delingette (EPIONE)	Inria, France
2023 - now	Ramya Muthukrishnan - PhD student Equivariant networks for 3D fetal head pose estimation in motion-aware MRI acquisition at high-resolution Co-supervised with Polina Golland (MIT)	MIT, USA
2024	Sebo Diaz - PhD student Landmark detection for fetal motion tracking in whole-uterus MRI time-series Co-supervised with Elfar Adalsteinsson (MIT) and Polina Golland (MIT)	MIT, USA
2019	Bo Hyun Song - MSc student Simulation of histological artefacts in medical images Co-supervised with Juan Eugenio Iglesias (UCL)	University College London, UK

Engagement with the scientific community

EVENT ORGANISATION

2023-2024	4th and 5th Boston medical Imaging Workshops Head of the organisation committee
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REVIEWING

Area Chair	MIDL (2024, 2025)	IEEE transactions on Medical Imaging, Medical Image Analysis, Nature Communications, Nature Biomedical Engineering, IEEE transactions on Pattern Analysis and Machine Intelligence, Scientific Data, NeuroImage, Imaging Neuroscience, Human Brain Mapping, Frontiers in Neuroscience, Journal of Neurology, Machine Learning for Biomedical Imaging
Reviews	Journals	
	Conferences	MICCAI (2021-2025), MIDL (2022, 2023), IPMI (2023), DGM4H NeurIPS Workshop (2023)
	Workshop proposals	EurIPS (2025)

PRESS

2023	PhysicsWorld press release <i>AI creates high-resolution brain images from low-field strength</i> by Cynthia E. Keen
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TEACHING

2018 **Introductory Mathematics for Computer Science** (10x1h30) University College London, UK

PANELLIST

2025 **Best practices for MICCAI reviews and rebuttals** MICCAI student board webinar, virtual
2024 **How to write an award-winning paper?** MIDL Winter Workshop, virtual

SOCIETY MEMBERSHIP

2020 – now **MICCAI member**

Invited presentations

Network conditioning for synergistic learning on partial supervision

202 Sabuncu Lab seminars Cornell Tech, New York City, USA
2024 LEMoN group lab seminars Harvard Medical School, Boston, USA

SE(3)-equivariant and noise-invariant 3D rigid motion tracking in brain MRI

2024 ARAMIS seminars Inria, Paris, France
 IMAGES seminars Télécom Paris, France
2023 EPIONE seminars Inria, Nice, France
 MICS seminars CentraleSupélec, Paris, France

Robust segmentation of heterogeneous clinical brain MRI scans

2025 Statlearn'25 spring school Inria, Nice, France
 3IA Côte d'Azur seminars 3IA, Nice, France
2024 Martinos center lab seminars Harvard Medical School, Boston, USA
 CMIC-WEISS seminars UCL, London, UK
2023 LEMoN group lab seminars Harvard Medical School, Boston, USA

SynthSeg: domain-randomisation for segmentation of brain MRI of any contrast and resolution

2022 Biomedical imaging and analysis seminars MIT, Boston, USA
2021 CMIC-WEISS seminars UCL, London, UK
2020 LCN group seminars Harvard Medical School, Boston, USA
 Bio-imaging symposium Imperial College London, UK

Publications

JOURNAL PAPERS

A probabilistic histological atlas of the human brain for MRI segmentation

Casamitjana, Mancini, Robinson, Peter, Annunziata, Althonayan, Crampsie, Blackburn, **Billot**, Atzeni, Puonti, Balbastre, Schimdt, Hughes, Augustinack, Edlow, Zollei, Thomas, Kliemann, Bochetta, Strand, Holton, Jaunmuktane, Iglesias
Nature (2025)

Benchmarking of deep learning methods for generic MRI multi-organ abdominal segmentation

Krishnaswamy, Ciaus, Pieper, Kikinis, **Billot**, Fedorov
Journal of Medical Imaging (2025)

A contrast-agnostic method for ultra-high resolution claustrum segmentation

Mauri, Fritz, Mora, **Billot**, Iglesias, Van Leemput, Augustinack, Greve
Human Brain Mapping (2025)

Synthetic data in generalizable, learning-based neuroimaging

Gopinath, Hoopes, Alexander, Arnold, Balbastre, **Billot**, Casamitjana, Cheng, Chua, Edlow, Fischl, Gazula, Hoffmann, Keene, Kim, Kimberly, Laguna, Larson, Van Leemput, Puonti, Rodrigues, Rosen, Tregidgo, Varadarajan, Young, Dalca, Iglesias
Imaging Neuroscience (2024)

Brain MRI signatures across sex and CSF Alzheimer's disease biomarkers

Cheng, He, Gopinath, **Billot**, Iglesias, Wu, Dodge, Wills, Carlyle, Kivisakk, Hyman, Arnold, Das
Brain Communications (2024)

SE(3)-equivariant and noise-invariant 3D rigid motion tracking in brain MRI

Billot, Dey, Moyer, Hoffmann, Abaci Turk, Gagoski, Grant, Golland
IEEE Transactions on Medical Imaging (2024)

Robust machine learning segmentation for large-scale analysis of heterogeneous clinical brain MRI datasets

Billot, Magdamo, Cheng, Arnold, Das, Iglesias
PNAS: Proceedings of the National Academy of Sciences (2023)

Linking brain structure, cognition, and sleep: insights from clinical data

Wei, Ganglberger, Sun, Hadar, Gollub, Pieper, **Billot**, Au, Iglesias, Cash, Kim, Shin, Westover, Thomas
Sleep (2023)

Brain Growth Charts for Quantitative Analysis of Pediatric Brain MRI Scans with Limited Imaging Pathology

Schabdach, Schmitt, Sotardi, Vossough, Andronikou, Roberts, Huang, Padmanabhan, Ortiz, Gardner, Covitz, Bedford, Mandal, White, Bullmore, Bethlehem, Shinohara, **Billot**, Iglesias, Ghosh, Gur, Satterthwaite, Roalf, Seidlitz, Alexander-Bloch
Radiology (2023)

Machine learning of dissection photographs and surface scanning for quantitative 3D neuropathology

Gazula, Tregidgo, **Billot**, Balbastre, William, Herisse, Deden, Casamitjana, Melief, Latimer, Kilgore, Montine, Robinson, Blackburn, Marshall, Connors, Oakley, Frosch, Young, Van Leemput, Dalca, Fischl, Mac Donald, Keene, Hyman, Iglesias
eLife (2023)

SynthSR: a public AI tool to turn heterogeneous clinical brain scans into high-resolution T1-weighted images for 3D morphometry

Iglesias, **Billot**, Balbastre, Magdamo, Arnold, Das, Edlow, Alexander, Golland, Fischl
Science Advances (2023)

SynthSeg: segmentation of brain MRI scans of any contrast and resolution without retraining

Billot, Greve, Puonti, Thielscher, Van Leemput, Fischl, Dalca, Iglesias
Medical Image Analysis (2023)

Quantitative brain morphometry of portable low-field-strength MRI using super-resolution machine learning

Iglesias, Schleicher, Laguna, **Billot**, Schaefer, McKaig, Goldstein, Sheth, Rosen, Kimberly
Radiology (2022)

In vivo hypothalamic regional volumetry across the frontotemporal dementia spectrum

Shapiro, Todd, **Billot**, Cash, Iglesias, Warren, Rohrer, Bocchetta
NeuroImage Clinical (2022)

A deep learning toolbox for automatic segmentation of subcortical limbic structures from MRI images

Greve, **Billot**, Cordero, Hoopes, Hoffmann, Dalca, Fischl, Iglesias, Augustinack
NeuroImage (2021)

SynthMorph: learning contrast-invariant registration without acquired images

Hoffmann, **Billot**, Greve, Iglesias, Fischl, Dalca
IEEE Transactions on Medical Imaging (2021)

Joint super-resolution and synthesis of 1mm isotropic MPRAGE volumes from clinical MRI exams with scans of different orientation, resolution and contrast

Iglesias, **Billot**, Balbastre, Tabari, Conklin, Gonzalez, Alexander, Golland, Edlow, Fischl
NeuroImage (2021)

Automated segmentation of the hypothalamus and associated subunits in brain MRI

Billot, Bocchetta, Todd, Dalca, Rohrer, Iglesias
NeuroImage (2020)

PEER-REVIEWED CONFERENCE PROCEEDINGS

Spatial regularisation for improved accuracy and interpretability in keypoint-based registration

Billot, Muthukrishnan, Abaci-Turk, Grant, Ayache, Delingette, Golland
MICCAI: Medical Image Computing and Computer-Assisted Intervention (2025)

Fetal pose estimation across gestational ages via cross-population augmentation

Diaz, **Billot**, Dey, Zhang, Abaci Turk, Grant, Golland, Adalsteinsson
MICCAI: Medical Image Computing and Computer-Assisted Intervention (2025)

Fetuses made simple: modelling and tracking of fetal shape and pose

Liu, Wang, Diaz, Abaci Turk, **Billot**, Grant, Golland,
MICCAI: Medical Image Computing and Computer-Assisted Intervention (2025)
Early accept

Robust rigid MRI-TRUS registration in prostate cancer using attention-CNN and ICP

Kattel, **Billot**, Facente, Delingette, Ayache
ASMUS: Workshop of Advances in Simplifying Medical UltraSound (2025)
Oral presentation

Multi-stage CNN for fast registration of 3D preoperative CTs to 2D intraoperative X-rays

Facente, **Billot**, Gopalakrishnan, Kattel, Wei, Golland, Delingette, Ayache, Berthet-Rayne
COLAS: Collaborative Intelligence and Autonomy in Image-guided Surgery (2025)

Learning general-purpose biomedical volume representations using randomized synthesis

Dey, **Billot**, Wong, Wang, Ren, Grant, Dalca, Golland
ICLR: International Conference on Learning Representations (2025)

Network conditioning for synergistic learning on partial annotations

Billot, Dey, Abaci Turk, Grant, Golland
MIDL: Medical Image with Deep Learning (2024)
Short-listed for best paper award, oral presentation

AnyStar: domain randomized universal star-convex 3D instance segmentation

Dey, Abulnaga, **Billot**, Abaci Turk, Grant, Dalca, Golland
WCACV: Winter Conference on Applications of Computer Vision (2024)
Early accept

Association of brain MRI signatures with Alzheimer disease CSF biomarkers

Cheng, Gopinath, **Billot**, Iglesias, Touroutoglou, Kivisäkk, Wu, Dodge, Hyman, Arnold, Das
Alzheimer's Association International Conference (2024)

Quantifying white matter hyperintensity and brain volumes in heterogeneous clinical and low-field portable MRI

Laso, Cerri, Sorby, Guo, Mateen, Goebel, Wu, Liu, Li, Young, **Billot**, Puonti, Sze, Sheth, Rosen, Kirsch, Strisciuglio, Wolterink, Eshaghi,
Barkhof, Kimberly, Iglesias
ISBI: International Symposium on Biomedical Imaging (2024)

Domain-agnostic segmentation of thalamic nuclei from joint structural and diffusion MRI

Tregidgo, Soskic, Olchanyi, Althonayan, **Billot**, Maffei, Golland, Yendiki, Alexander, Bocchetta, Rohrer, Iglesias
MICCAI: Medical Image Computing and Computer-Assisted Intervention (2023)
Early accept

Equivariant and denoising CNNs to decouple intensity and spatial features for motion tracking in fetal brain MRI

Billot, Moyer, Karani, Hoffmann, Abaci Turk, Grant, Golland
MIDL: Medical Image with Deep Learning (2023), short paper track

Robust segmentation of brain MRI in the wild with hierarchical CNNs and no retraining

Billot, Magdamo, Arnold, Das, Iglesias
MICCAI: Medical Image Computing and Computer-Assisted Intervention (2022)

Super-resolution of portable low-field MRI in real scenarios: integration with denoising and domain adaptation

Laguna, Schleicher, **Billot**, Schaefer, McKaig, Goldstein, Sheth, Rosen, Kimberly, Iglesias
MIDL: Medical Image with Deep Learning, short paper track (2022)

Joint segmentation of multiple sclerosis lesions and brain anatomy in MRI scans of any contrast and resolution

Billot, Cerri, Van Leemput, Dalca, Iglesias
ISBI: International Symposium on Biomedical Imaging (2021)
Oral presentation

Learning MRI contrast-agnostic registration

Hoffmann, **Billot**, Iglesias, Fischl, Dalca
ISBI: International Symposium on Biomedical Imaging (2021)
Oral presentation

Partial volume segmentation of brain MRI scans of any contrast and resolution

Billot, Robinson, Dalca, Iglesias
MICCAI: Medical Image Computing and Computer-Assisted Intervention (2020)
Oral presentation, early accept

A learning strategy for contrast-agnostic MRI segmentation

Billot, Greve, Van Leemput, Fischl, Dalca, Iglesias

MIDL: Medical Image with Deep Learning (2020)

Short-listed for best paper award, oral presentation

Image synthesis with a convolutional capsule generative adversarial network

Bass, Dai, **Billot**, Arulkumaran, Creswell, Clopath, De Paola, Bharath

MIDL: Medical Image with Deep Learning (2019)

Short-listed for best paper award, oral presentation

Deep reinforcement learning for subpixel neural tracking

Dai, Dubois, Arulkumaran, Campbell, Bass, **Billot**, Uslu, De Paola, Clopath, Bharath

MIDL: Medical Image with Deep Learning (2019)

Spotlight

CONFERENCE ABSTRACTS

Network conditioning for synergistic learning on partial annotations

Billot, Dey, Abaci Turk, Grant, Golland

Colloque Français d'Intelligence Artificielle en Imagerie Biomédicale (2025)

Physics-based generative models in low data regime: application to segmentation of cortical microscopies

Billot, Bass, Bharath

Human Brain Project 4th summer school, Obergurgl, Austria (2017)

Oral presentation